# EVERY OMEGA-TREE WITH FINITE LEVELS CONTAINS A LEAFLESS BALANCED SUBTREE WITH OMEGA LEVELS

SUPAKUN PANASAWATWONG AND J K TRUSS AND SUPAKUN PANASAWATWONG, J. K. TRUSS

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*Allowed standard prerequisites (editorial).*

- The notion of a balanced subtree and of the degree of a node on a level. Defined in the paragraphs preceding the counted statement (the discussion ending at source line 489). Fixes what has to be produced and what the pruning at each stage must achieve.
- A sequence of natural numbers in which every number occurs infinitely often. Used inside the counted proof at lines 494-495 and 508. Supplies the schedule according to which the successive prunings are performed and why each level is handled infinitely often.

**Lemma 0.1.** *Any tree $(T, \leq)$ with $\omega$ levels, all finite, has a balanced subtree, having no leaves, and also having $\omega$ levels.*

*Proof.* Let $S$ be a sequence of natural numbers such that every number occurs infinitely often, say $S = \langle k_n : n \in \omega \rangle$. We construct a decreasing sequence $T_n$ of subtrees of $T$, each having $\omega$ levels, such that for each $n$, $T_n$ is pruned on level $k_n$ so that every member has the same degree (number of immediate successors) on that level. Let $L_n$ be the $n$th level of $T$.

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First let $T_0 = T$. Now suppose that $T_n$ has been constructed. Let $X_n$ be the set of members $x$ of height $k_n$ in $T_n$ such that $\{y \in T_n : x \leq y\}$ is infinite, and subject to this have minimum degree (since $T_n$ has $\omega$ levels, at least one node on that level has infinitely many points above it). Let $T_{n+1}$ be the subtree obtained by removing all members of $T_n \geq$ some member of $L_{k_n} \setminus X_n$. Then $T_{n+1}$ is a subtree of $T_n$ also having $\omega$ levels. Let $T^* = \bigcap_{n\in\omega} T_n$. It remains to show that $T^*$ is balanced and has no leaves.

Consider any $n$. Since each member of $\omega$ is listed infinitely often, $\{m : k_m = n\}$ is infinite. Let $L^m_n$ be the $n$th level of $T_m$. Then $L^0_n \supseteq L^1_n \supseteq L^2_n \ldots$ is a decreasing sequence of non-empty finite sets, so is eventually constant, and clearly this equals the $n$th level of $T^*$. Furthermore every $x$ in the $n$th level of $T^*$ has at least one successor in every $T_m$, and hence also in $T^*$, so is not a leaf, and for the same reason, its degree is constant among nodes on that level. Hence $T^*$ is a balanced subtree of $T$. □